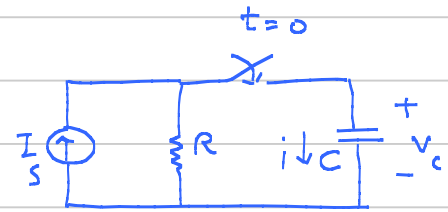




The step response of an RC circuit



Apply KCL: $\frac{V_c}{R} + C \frac{dV_c}{dt} = I_s$

Similar to the derivation of the step response of RL circuit:

$$V_c = I_s R + (V_0 - I_s R) e^{-\frac{t}{\tau}}, \quad \tau = RC$$

$\uparrow$ final value $\uparrow$ initial value

General solution for step and natural responses

For computing the step or natural response of RL and RC circuits follow these steps:

1. Identify the variable of interest. Choose inductor current in an RL and capacitor voltage in an RC circuit.
2. Determine the initial value of the variable using the circuit before switching when all inductors are short and capacitors are open at steady state.
3. Determine the final value of the variable using the circuit after switching when all inductors are short and capacitors are open at steady state.
4. Calculate the time constant by finding the equivalent resistance connected to the inductor or capacitor in the circuit after switching. ($\tau = L/R$ in an RL circuit and $\tau = RC$ in an RC circuit)
5. Use this general equation to find the expression for the variable:

$$x(t) = x_f + [x_0 - x_f] e^{-\frac{t}{\tau}}$$

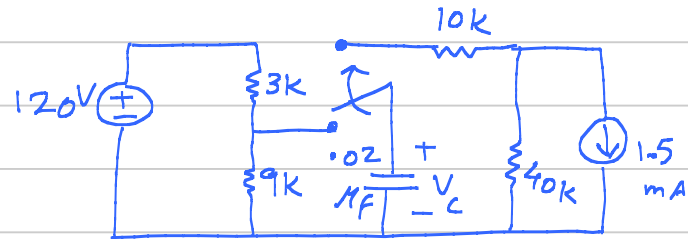
$x(t)$: unknown variable (i_L or V_c)

x_f : final value, τ : time constant ($\frac{L}{R}$ or RC)

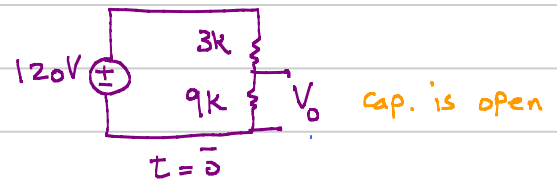
$x(t_0)$: initial value

Ex. Find a) $v_c(0)$, b) $v_c(\infty)$, c) τ ,

d) the time required for the capacitor voltage to reach zero after switching.

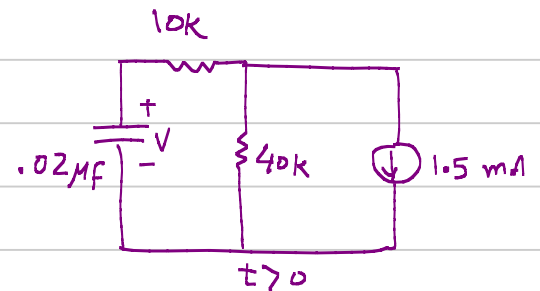


$$V_o = 120 \frac{9}{9+3} = \underline{90 \text{ V}}$$



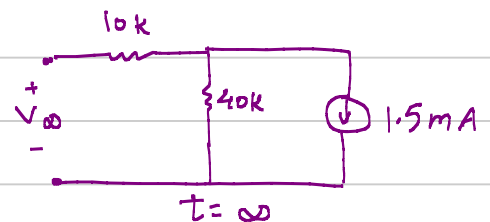
$$\tau = R_{eq} \cdot C = (10k + 40k)(0.02 \mu) = \underline{1ms}$$

$$V_\infty = -(40k)(1.5mA) = \underline{-60 \text{ V}}$$



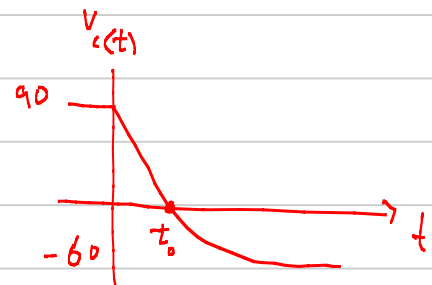
$$V_c(t) = -60 + (90 - (-60)) e^{-\frac{t}{1ms}}$$

$$V_c(t) = -60 + 150 e^{-1000t} \text{ V}$$



$$V_c(t) = 0 \rightarrow 150 e^{-1000t_0} = 60$$

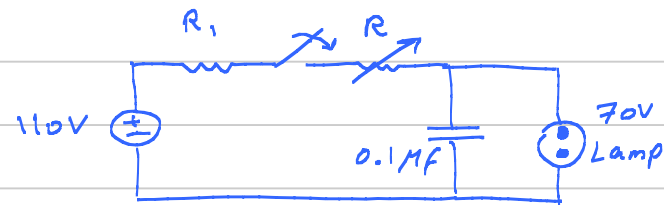
$$\underline{t_0 = 916.3 \mu s}$$



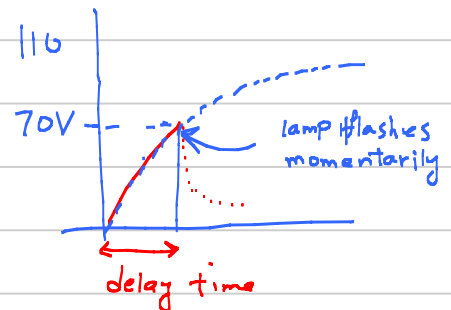
Practical Example, Delay Circuit

This circuit is an RC circuit that provides various time delays.

when the switch is closed, the capacitor voltage increases toward 110V.



when it reaches 70V, the lamp conducts. Because of the low resistance of the lamp, the time constant decreases and the capacitor voltage drops fast and the lamp goes off.



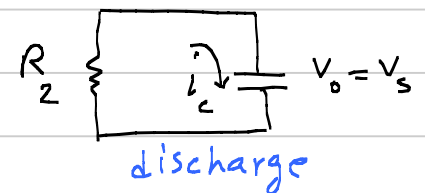
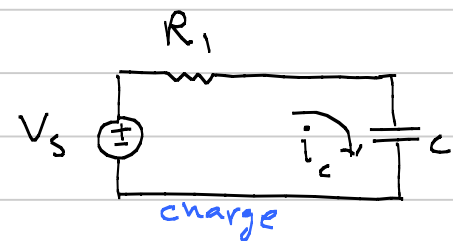
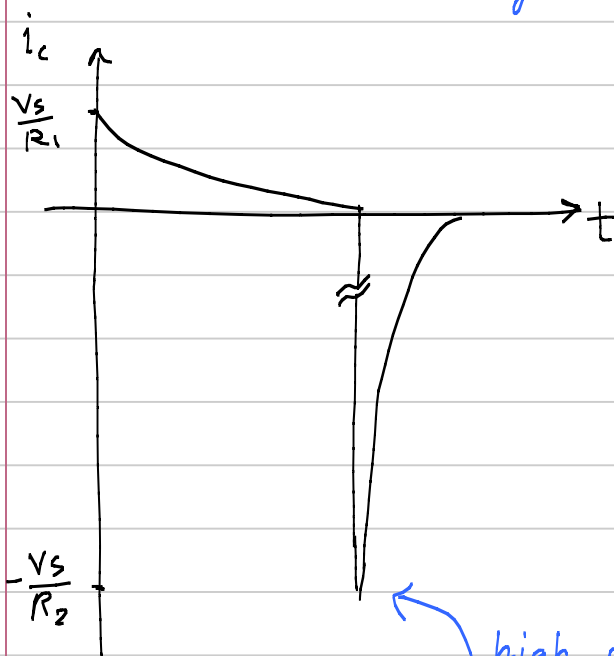
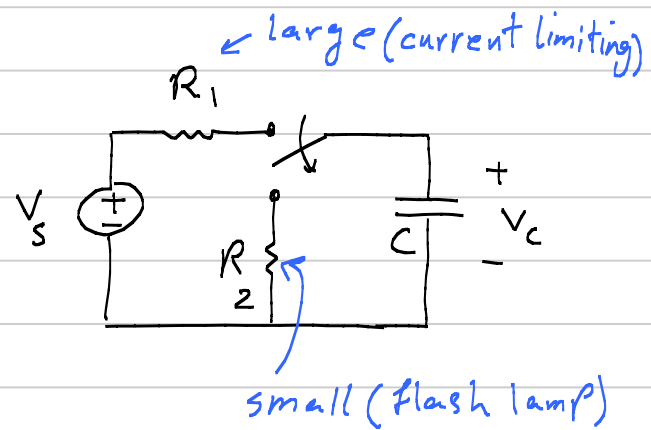
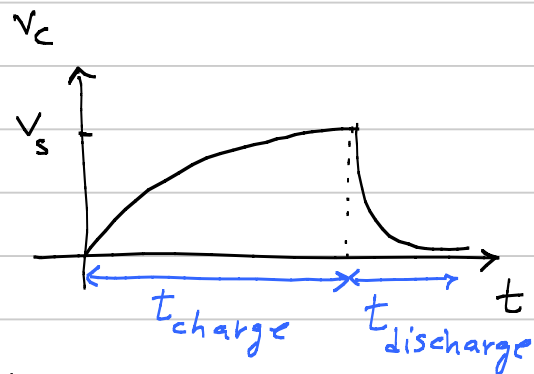
$$\tau_{\text{charge}} = (R + R_1) C, \quad (\text{Lamp off})$$

$$\tau_{\text{discharge}} = \left[(R_1 + R) \parallel R_{\text{lamp}} \right] C, \quad (\text{Lamp on})$$

by adjusting R , the time constant is changed

and the delay time before the lamp flashes, is changed.

Practical Example, Photo Flash



high discharge current with the peak of $\frac{V_s}{R_2}$ for a short duration.

If $V_s = 80 \text{ V}$, $C = 2 \text{ mF}$, $R_1 = 2 \text{ k}\Omega$, $R_2 = 100 \Omega$, find charge time, discharge time and peak flash current.

$$t_{\text{charge}} = 5\tau_1 = 5R_1C = 5(2000)(2 \times 10^{-3}) = \underline{20 \text{ sec.}}$$

$$t_{\text{discharge}} = 5\tau_2 = 5R_2C = 5(100)(2 \times 10^{-3}) = \underline{1 \text{ sec.}}$$

$$i_{\text{pk}} = \frac{V_s}{R_2} = \frac{80}{100} = \underline{800 \text{ mA}}$$